Google Cloud Skills Boost Cert.
Introduction to AI and Applications Courses Work

- What problems will the model solve?
- Who are the intended users?
- What other groups may be impacted?
- What groups are invisible today?
- How was the training data collected, sampled and labeled?
- Is the training data skewed?
- How was the model tested and validated?
- Is the model behaving as expected?

Used Material:

This AI model is helping researchers detect disease based on coughs

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2 min read

Here's how our bioacoustic foundation model, Health Acoustic Representations (HeAR), may help improve health outcomes for people across India.

Shravya Shetty
Google Research Director of Engineering

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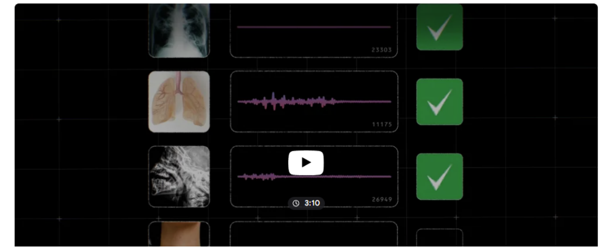

Google Cloud Skills Boost Courses Work

Topic: AI model is helping researchers detect disease based on coughs

(High Concept: Use of The ethical aims of Google's AI principles)

*Select four topics to answer.

Q1. What problems will the model solve?

The model analyses the key features in each cough and respiration (breathing) sound produced by specific diseases. It also helps researchers understand what causes these sounds and their underlying causes.

Q2, Q3. Who are the intended users? What other groups may be impacted?

The intended users are **experienced medical professionals and research teams under respiration speciality at first**.

Moreover, in **medical education with interactive media**, developers may find it helpful when **developing educational content** from this project. It is because some specific diseases in respiration are contagious, and it is not to hoist the risk so high by visiting a contaminated environment for educational purposes.

Q4. How was the training data collected, sampled and labelled?

When the cough tracks are gathered, they should be uploaded and stored into a platform or control system internally accessed by the user (person who coughs) and research teams.

The **tracks are anonymised**, with the help of **unique random identifiers** and a **serialisation process that makes the sample not contain personal data from (coughing) users.**

However, a control number that **is blind to the research team but occultate to the user** should be assigned to each sample of coughs if follow-up action is needed, just like the COVID-19 PCR testing result.